

# Kunal Maheshwari

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## Summary

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**Backend Software Engineer / SDE-2** with 4.6+ years building and modernizing enterprise Java systems across healthcare integrations and automotive products, including services handling **50K+ daily requests under a 99.99% availability SLA**.

**Delivery scope:** Takes backend changes from requirement analysis through Java/Spring implementation, REST/SOAP and database integration, testing, release validation, and production support; extends this foundation through project-based AWS data pipelines.

**Engineering approach:** Improves reliability, performance, security-boundary handling, and observability through explicit failure paths, correlation telemetry, set-based query optimization, and structured data flows.

**Working style:** Owns deliverables independently, resolves ambiguity through focused analysis, and collaborates with product-facing, engineering, QA, database, deployment, and vendor teams while documenting decisions, reviewing code, and mentoring developers.

## Technical Skills

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**Java Backend:** Java 8/11, Spring Boot (knowledge), Spring Framework/MVC, REST APIs, SOAP, JAX-WS, Java EE/J2EE, Hibernate/HQL, JDBC/OJDBC, SOA, design patterns.

**Databases & Integration:** SQL, Oracle, DB2, PostgreSQL (projects/training), query optimization; JMS, Azure Service Bus, Azure Relay, Azure Functions, IBM WebSphere, Oracle WebLogic.

**Cloud & Data Projects:** AWS S3, Lambda, EFS, ECS, RDS, CloudWatch, IAM, VPC; Docker, Terraform, Spark/PySpark, Spark Structured Streaming, PostgreSQL/JDBC, Flask/Plotly analytics.

**Security, Quality & Delivery:** API Gateway security boundary, service authorization, Git, Maven, Jenkins, GitLab CI/CD, IBM UrbanCode, JUnit/JUnit4, integration testing, SoapUI, Postman, Log4j2, Agile/Scrum.

## Work Experience

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**NTT Data**, Bengaluru, India

*Software Developer, Aug 2022 - Aug 2024*

- Developed and maintained **Java 8/11, JAX-WS/Spring, and REST/SOAP** healthcare integration services on IBM WebSphere, supporting **50K+ daily requests under a 99.99% availability SLA**.
- Integrated synchronous REST dependencies into established SOAP/SOA workflows with explicit timeout and partial-failure handling across member and provider data flows.
- Connected Java services to Oracle views and PL/SQL procedures through OJDBC, implementing service-layer business rules while preserving database ownership boundaries.
- Integrated flows across Azure Service Bus, Azure Relay, Azure Functions, and JMS; consumed authenticated caller identity and enforced service authorization behind an enterprise API Gateway boundary.
- **Led a project-wide Log4j2 migration**, owning the initial PoC, preserving production logging contracts, documenting a repeatable upgrade path, and reviewing the team rollout.
- Integrated AOP-based monitoring that propagated correlation IDs and captured latency, success/failure, and request-volume telemetry across service calls.
- Supported root-cause analysis of a WebSphere memory/resource-retention incident through code-path review, mitigation planning, and rolling-restart validation while service availability was maintained.
- Supported Maven/Jenkins/IBM UrbanCode delivery, JUnit/SoapUI testing, release documentation, post-deployment validation, and go/no-go or rollback recommendations.

Capgemini, Mumbai, India

Senior Analyst, Jul 2020 - Aug 2022

- Developed and supported a **multi-tier Java EE/J2EE, JSF/AJAX, Hibernate/HQL** application on Oracle WebLogic with DB2 for Mercedes/Daimler Vans configuration and recommendation workflows.
- Integrated a Python prediction service through asynchronous Java REST requests and callbacks, then applied deterministic option and package validation before recommendations reached dealer users.
- **Reduced VIN repeat-configuration API latency from 60+ seconds to 10–20 seconds** through set-based HQL/DB2 processing while retaining authoritative business validation in Java.
- Diagnosed failures across Python-service output, Java validation logs, and DB2 reference data; supported Jenkins/GitLab CI/CD, JUnit4, Git-based reviews, and mentored developers through features and production defects.

Capgemini, Mumbai, India

Analyst Trainee, Jan 2020 - Jun 2020

- Built Java Spring/Spring MVC, Angular, and PostgreSQL modules for a **digital-wallet training application**, implementing authentication, REST-style workflows, account balances, transactions, and profile features.

## Selected Projects

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### AWS Automated Spark Pipeline

[github.com/preferablehuman/AWS-automated-spark-pipeline](https://github.com/preferablehuman/AWS-automated-spark-pipeline)

- Built an event-driven data pipeline where **S3 uploads trigger Lambda staging to EFS**, an ECS-hosted Docker/Spark Structured Streaming workload processes micro-batches, and results persist to RDS PostgreSQL/JDBC.
- Provisioned VPC, IAM, Lambda, EFS, ECS, RDS, security groups, and CloudWatch with **Terraform**; processed an **8GB+ dataset**, retained checkpoints for recovery, and exposed processed data through Flask/Plotly analytics views.

### CodingHelper - AI Coding Platform

[github.com/preferablehuman](https://github.com/preferablehuman)

- Built a project-based LangChain/Ollama application with **PostgreSQL as the source of truth**, RAG/Qdrant retrieval, isolated Java/Python execution, and bounded asserting-test gates before generated solutions were reused.

## Education

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Central Michigan University, Mount Pleasant, MI, USA

Aug 2024 - Dec 2025

Master of Science in Computer Science, GPA: 3.87/4.0

Lovely Professional University, Punjab, India

Aug 2016 - May 2020

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.02/10

## Certifications

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Microsoft Certified: Azure Fundamentals | Azure Developer Associate | Azure Data Engineer Associate